

MORRASALE, Russian FederationWMO: **230320**Lat: **69.714N**Lon: **66.814E**Elev: **86**StdP: **14.65**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-36.8	-32.5	-41.3	0.5	-36.9	-37.1	0.7	-32.4	43.4	2.5	37.7	1.1	12.1	60	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	67.9	61.1	63.1	57.5	58.4	54.3	61.6	67.1	58.0	62.2	54.6	58.0	11.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	74.3	65.3	55.6	66.1	61.1	52.4	58.5	56.5	27.6	67.1	25.2	62.2	23.0	58.1	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.9	31.7	28.1	-41.6	73.2	6.4	6.2	-46.2	77.6	-49.9	81.2	-53.5	84.6	-58.1	89.1		
(5)			-39.9	63.8	7.4	4.0	-45.2	66.7	-49.6	69.0	-53.7	71.2	-59.1	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	-3.5	-5.2	2.4	10.9	23.4	38.0	47.2	45.6	39.7	27.6	10.6	3.6		
	DBStd	22.04	15.35	16.93	15.13	12.52	8.37	7.74	9.57	6.03	5.77	9.69	13.30	16.38		
	HDD50	11017	1660	1545	1474	1173	823	373	174	165	313	695	1183	1439		
	HDD65	16367	2125	1965	1939	1623	1288	810	559	601	760	1160	1633	1904		
	CDD50	132	0	0	0	0	0	13	87	29	3	0	0	0		
	CDD65	7	0	0	0	0	0	0	7	0	0	0	0	0		
	CDH74	27	0	0	0	0	0	0	27	0	0	0	0	0		
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0		
	WSAvg	14.1	14.7	14.3	14.4	14.2	14.0	12.9	11.9	12.5	13.3	15.3	15.7	15.5		
	PrecAvg	10.1	0.6	0.5	0.4	0.5	0.6	1.0	1.2	1.7	1.7	1.0	0.6	0.6		
	PrecMax	17.2	1.8	1.2	1.3	1.4	1.8	2.3	2.2	3.8	3.6	2.2	1.5	1.7		
	PrecMin	5.5	0.0	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.2	0.4	0.2	0.2		
	PrecStd	3.1	0.4	0.3	0.3	0.3	0.4	0.5	0.6	0.9	0.9	0.5	0.4	0.4		
	DB	30.0	29.6	31.7	35.6	46.0	67.4	77.6	66.9	56.2	43.9	34.0	32.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% MCWB	29.4	29.0	30.9	33.5	43.1	58.3	64.5	60.8	53.1	42.5	33.1	31.2			
	2% DB	26.6	26.2	29.7	33.3	39.2	60.6	71.4	61.7	52.8	42.1	31.8	30.0			
	2% MCWB	26.2	25.6	29.2	32.3	37.2	53.7	63.8	57.4	50.8	40.9	31.0	29.2			
	5% DB	22.2	23.0	27.5	32.1	35.8	55.4	66.8	57.8	50.0	40.5	30.1	27.9			
	5% MCWB	21.5	22.4	27.0	31.5	34.5	50.4	60.4	54.8	48.3	39.5	29.4	27.2			
	10% DB	18.1	18.4	24.3	29.8	33.6	50.1	62.3	54.3	47.6	38.8	28.0	25.5			
	10% MCWB	17.4	17.8	23.7	28.9	32.7	46.8	57.1	52.0	46.2	37.9	27.3	24.9			
	0.4% WB	29.5	29.1	31.2	33.8	42.8	58.5	66.4	62.0	54.0	43.1	33.3	31.4			
	0.4% MCDB	29.8	29.4	31.4	34.6	44.8	65.5	73.9	65.9	55.7	43.7	33.5	31.7			
	2% WB	26.0	25.6	29.2	32.6	37.2	54.3	63.8	58.0	51.0	41.1	31.3	29.4			
	2% MCDB	26.4	26.0	29.6	33.2	38.7	60.1	70.7	61.0	52.4	41.8	31.7	29.9			
	5% WB	21.6	22.4	26.9	31.6	34.6	50.5	61.0	55.1	48.6	39.5	29.5	27.3			
	5% MCDB	22.1	22.9	27.5	32.2	35.5	55.0	66.5	57.5	49.8	40.2	30.1	27.8			
	10% WB	17.4	17.7	23.8	29.0	32.9	46.7	57.5	52.3	46.3	38.0	27.4	24.9			
	10% MCDB	18.1	18.3	24.3	29.8	33.6	50.0	61.8	54.1	47.5	38.8	28.0	25.5			
(35) Mean Daily Temperature Range	MDBR	12.6	11.9	12.4	12.4	8.3	9.1	10.9	7.6	6.1	6.9	11.4	12.6			
	5% DB MCDBR	13.7	13.5	11.1	10.7	8.7	18.7	17.4	14.8	8.9	5.0	8.5	11.5			
	5% DB MCWBR	13.7	13.1	11.2	10.4	7.6	13.2	10.6	10.7	7.0	4.8	8.8	11.5			
	5% WB MCDBR	13.8	13.2	11.0	10.4	8.1	18.1	17.6	13.5	8.0	4.9	8.5	11.5			
	5% WB MCWBR	13.8	13.2	11.1	10.2	7.2	13.1	11.7	10.4	6.6	5.1	8.9	11.5			
(40) Clear-Sky Solar Irradiance	taub	0.205	0.239	0.290	0.316	0.330	0.342	0.370	0.355	0.313	0.252	N/A	N/A			
	taud	1.853	1.960	2.022	2.039	2.099	2.288	2.372	2.431	2.458	2.177	N/A	N/A			
	Ebn at Noon	3	148	220	255	267	269	256	246	225	148	0	0			
	Edh at Noon	1	14	28	37	40	34	31	26	19	11	0	0			
(44) All-Sky Solar Radiation	RadAvg	3	107	609	1299	1862	1899	1610	984	493	134	0	0			
	RadStd	0	20	49	71	95	153	217	110	41	19	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+3.10	+5.00	+6.05	+2.95	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page